SESSION 4 LECTURE NOTES

Advanced Data Fusion in Google Earth Engine: Architectural Principles from Raw Sensors to 4D Feature Cubes

GeoAI Fusion Bootcamp 2026

Introduction

Session 2 established the conceptual principles of data fusion, and this session translates those principles into a concrete, executable workflow using Google Earth Engine. Google Earth Engine combines petabyte-scale spatiotemporal data storage with cloud computation, making it possible to harmonise and fuse satellite datasets without downloading raw imagery to a local machine.

Single-sensor analysis fails for three connected reasons. Optical imagery cannot penetrate cloud cover or heavy smoke, a serious constraint in tropical or storm-prone regions. A single spectral band range cannot always separate materials that share an identical signature in that range but differ elsewhere, a limitation known as spectral ambiguity. A single satellite overpass, moreover, misses events that occurred days before or after it, failing to capture dynamic change. Treating any one image or dataset as complete truth inherits every one of these gaps.

This session's objective is to show participants how Google Earth Engine's application programming interface implements harmonisation, compositing, and band-level fusion in practice, culminating in the construction of a four-dimensional feature cube ready for machine learning.

Discussion

Data Harmonisation as a Prerequisite

Multi-sensor integration in Google Earth Engine depends on the same alignment rules introduced in Session 2. Consistent scales require resampling each dataset to match a target resolution, for instance aligning twenty-metre synthetic-aperture radar data to a ten-metre optical grid. Consistent projections require reprojecting every dataset to a common coordinate reference system for accurate spatial overlay. Temporal alignment requires aggregating observations to match exact time frames, for instance standardising daily climate data to a monthly composite. A harmonisation engine applies these three rules together, producing a unified, multi-layer data cube that shares the same resolution, projection, time frame, and spatial extent.

Compositing: Trading Temporal Detail for Spatial Clarity

A single daily satellite image is often sparse, patchy with cloud cover, and incomplete. Compositing addresses this problem much as a photographer stacks several exposures of a crowded street to remove passing pedestrians from the final image. Google Earth Engine converts a time series of images into a single, clean composite by masking clouds across the series and then applying a statistical reducer, such as the median

function, across the unmasked pixels. The result trades some temporal detail, since the composite no longer represents a single moment, for substantial gains in spatial completeness and clarity.

The Resampling Matrix

Aligning disparate pixel grids requires choosing among three resampling methods, and the choice depends on the nature of the underlying data. Nearest-neighbour resampling assigns the value of the closest input pixel and changes no data values, making it essential for categorical data, such as land-cover classes, since averaging class codes would produce meaningless numbers. Bilinear interpolation uses four neighbouring pixels to calculate a weighted average, producing a smoother output appropriate for continuous data, such as environmental variables or optical bands. Bicubic interpolation extends the averaging window to sixteen pixels, fitting a smooth curve and delivering the highest visual quality, though at a greater computational cost, making it best suited to visualisation rather than large-scale batch processing.

The Google Earth Engine Merging Toolkit

Google Earth Engine offers three distinct operations for connecting image geometries, and each serves a different fusion purpose. The merge function for image collections concatenates two collections into one along the temporal axis, for instance combining Landsat 8 and Landsat 9, provided the two collections share comparable band structures and units. The mosaic function, together with statistical reducers, resolves overlapping pixels captured in the same space. The mosaic function itself quickly keeps the top pixel from the stack, while a reducer such as the median function blends overlapping pixels statistically for time-series compositing. The addBands function performs the core operation of data fusion. It stacks bands from a second image onto a first image along the spectral axis, for instance attaching a radar band to an optical image, provided both images share the exact same projection and pixel grid.

Constructing the 4D Feature Cube

The synthesis stage combines three dimensions into a single structure. Spatial dimensions are harmonised through reprojection and resampling, ensuring every layer shares the same coordinate reference system and resolution. The spectral dimension stacks fused variables from diverse sensors together using the addBands function, for instance combining an optical band, a radar band, an elevation band, and a land-cover band into a single multi-band image. The temporal dimension aggregates observations via composites, representing specific epochs or seasons rather than single acquisition dates. The result is an organised, scalable, artificial-intelligence-ready dataset in which every pixel column contains a rich, multi-sensor mathematical vector.

Executing the Pipeline

A practical Google Earth Engine script illustrates the entire process concretely. Since synthetic-aperture radar data is continuous, the script applies bilinear interpolation to ensure a smooth mathematical transition onto the optical grid. The script then harmonises the radar composite by resampling and reprojecting it, forcing it to adopt the exact coordinate reference system of the optical band through a reproject call. Finally, the script fuses the two layers into a single feature cube using the addBands function, attaching the aligned radar bands onto the optical composite. This short sequence of harmonisation followed by fusion recurs throughout every advanced Google Earth Engine workflow in this bootcamp.

Conclusion

This session translated the conceptual principles of data fusion into a working Google Earth Engine pipeline. Harmonisation through resampling and reprojection, compositing through cloud masking and statistical reducers, and fusion through the addBands function together produce the four-dimensional feature cube that machine-learning models require.

Participants should now be able to write a short Google Earth Engine script that harmonises two sensor sources and fuses them into a single multi-band image. Session 5 builds on this technical foundation by examining the cloud infrastructure that scales these workflows from a single script to a production system.